

SCIBENCH ANONYMIZED REPLICATION DOSSIER

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Anonymous replication dossier: deterministic Floquet system identification

Scientific objective

Reproduce a deterministic stroboscopic Floquet-DMD experiment for a driven two-level quantum system. Learn a one-drive-period propagator from four training periods and extrapolate the Pauli-vector trajectory. No measurement noise or random sampling is used.

State dynamics

Use the Hamiltonian

[EQUATION]

$$H(t) = \pi \sigma_3 + \cos(2 \pi \omega_D t) \sigma_1, \\ \omega_D = 1.1.$$

The intrinsic period is 1 and the drive period is $T=1/\omega_D$. Start in the lower σ_3 eigenstate and record

[EQUATION]

$$x(t) = (\langle \sigma_1 \rangle, \langle \sigma_2 \rangle, \langle \sigma_3 \rangle)^T.$$

Equivalently, evolve the density matrix with the von Neumann equation or the real Bloch vector with

[EQUATION]

$$\begin{aligned} dx_1/dt &= -2 \pi x_2, \\ dx_2/dt &= 2 \pi x_1 - 2 \cos(2 \pi \omega_D t) x_3, \\ dx_3/dt &= 2 \cos(2 \pi \omega_D t) x_2. \end{aligned}$$

Floquet snapshots

Sample four equally spaced phases per drive period. Arrange one period in each column:

[EQUATION]

$$z_n = [x(nT); x(nT+T/4); x(nT+T/2); x(nT+3T/4)].$$

The one-period propagator exists because

[EQUATION]

$$P_T(t_0) = \exp(-i T H_F(t_0)).$$

Use z_0, z_1, z_2, z_3 as the four training columns. Apply rank-three exact DMD to the three offset pairs:

[EQUATION]

$$X = [z_0, z_1, z_2],$$

$$X' = [z_1, z_2, z_3].$$

If $X=U \Sigma V^\dagger$, diagonalize

[EQUATION]

$$A_{\text{tilde}} = U^\dagger X' V \Sigma^{-1}.$$

Return its three eigenvalues. Use the full fifth-period column z_4 as the initial condition and extrapolate eight subsequent periods. Submit the 32 predicted phase samples for periods beginning at $5T$ through $13T$ (endpoint excluded); do not include z_4 itself in the submitted prediction.

Output requirements

Save the eigenvalues as 'floquet_eigenvalues', a float array of shape (3,2) whose columns are real and imaginary parts. Save the 32 sample times as 'prediction_times' and the three Pauli components as 'prediction_trajectory' with shape (3,32). Eigenvalue order is arbitrary.

[MASKED: all eigenvalues, extrapolated values, comparisons, and original graphics are withheld.]